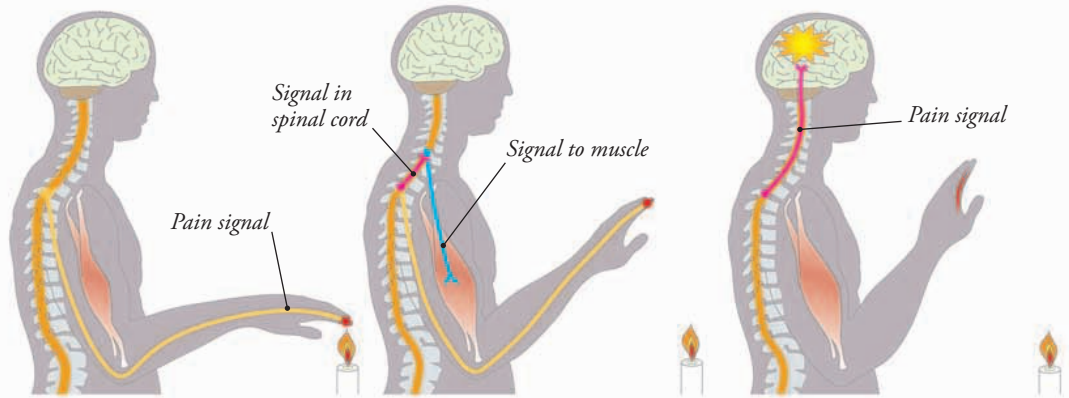


REFLEX ACTIONS

Most of the signals from your sense organs are processed by your brain before your body reacts. A reflex action, however, happens much more quickly since it takes a shortcut through your spinal cord, bypassing your brain. For example, if your hand gets too close to something hot, a reflex action immediately causes the muscle in your arm to pull your hand out of harm's way. This happens before your brain even has time to feel heat or pain.



▲ DANGER

Pain receptors in the skin detect the heat of a candle flame and send a signal to the spinal cord.

▲ WITHDRAWAL

The signal passes through the spinal cord and then travels to an arm muscle, which pulls the hand out of harm's way.

▲ PAIN

The pain signal reaches the brain. Only then does a person feel pain, after the hand has already pulled back.

PINS AND NEEDLES

If you lean on your elbow for a while, your arm might feel numb. This happens because the pressure cuts off the blood supply to your nerves, preventing them from sending touch messages to your brain. Stop leaning, and as the blood supply returns to your arm, you'll feel a tingling sensation called "pins and needles."



BABY REFLEXES

Babies are born with some built-in reflexes that fade during their first year. For example, a baby can swim underwater because a reflex stops water from getting into the lungs. Other baby reflexes include gripping when the hand is touched, and sucking to get milk when a nipple is put in the mouth.



GAG REFLEX

If something other than food touches the back of your throat, it triggers the gag reflex. This automatic squeezing of your throat stops unwanted objects from going any farther and prevents choking. Sword swallows learn to overcome the gag reflex so they can push a sword down their throat. About one-third of people have a weak gag reflex or no gag reflex.

